Grant Ho

(415) 812-8530 | grantho701@gmail.com | linkedin.com/in/grant-ho-85729b278/ | github.com/hihitherethere

Education

University of California, Los Angeles — Los Angeles, CA
BS in Computer Science — *Expected June 2029*

Relevant Courses:

- Differential Equations, Linear Algebra, Multivariable Calculus

Experience

Machine Learning Research Intern — Jun. - Sep. 2025
Aspiring Scholars Directed Research Program, Fremont, CA | PyTorch, Tensorflow, Numpy

- Predicted chaotic dynamical systems over long time horizons with $< 5\%$ error using neural networks
- Developed and evaluated 5+ neural architectures, including LSTM, GRU, and Reservoir computing
- Evaluated models using divergence time, F-statistics, R^2, and other statistical measures
- Applied neural forecasting methods to chaotic blood-dynamics models, drafted research paper with findings

Machine Learning Research Intern — Jun. - Sep. 2024
Aspiring Scholars Directed Research Program, Fremont, CA | SkLearn, Pandas, Scipy, Matplotlib

- Developed custom self-organizing map (SOM) to project high-dimensional data into a 2D topological representation
- Evaluated SOM performance with high-dimensional topological structures generated using parametric equations
- Integrated Python libraries sklearn, scipy, and numpy to build the model on Google Colab
- Created graphics visualizing data and results using Python library Matplotlib

Projects

Codeforces Difficulty Prediction Model | *Python, Flask, SQL, HTML, CSS, JS* — November 2024 - May 2025

- Predicted problem difficulty with a margin of error less than 10% when compared to real Codeforces data
- Implemented statistical regression techniques on old contest data to create difficulty predicting model
- Developed a full-stack web app using Flask and Supabase SQL database as backend, HTML/CSS/JS as frontend
- Automatically scrapes for new contest data using Codeforces API, ∼100 unique users

AI Coliseum 2024 – 5th place | *Java* — September 2024 - October 2024

- Designed symmetry-based exploration strategy using bot dispersion to efficiently map unknown game environment
- Implemented inter-agent communication algorithm to relay discovered map regions to a central base
- Developed a bugnav algorithm to navigate around walls and environmental hazards

Logicdle | *Python, Claude Code* — September 2026

- Utilized Claude Code to build, test, and deploy full-stack application in under 40 hours
- Implemented user submission pipeline, image storage and retrieval system, and leaderboard systems
- Reviewed code to ensure secure handling of confidential data such as passwords and database keys

Awards

Codeforces Master: Top 2.5% globally
USACO Platinum: Top 250 U.S. High Schoolers
4x AIME Qualifier
PicoCTF (Cybersecurity Competition): Top 100 in the High School division out of 2,000 teams

Technical Skills

Languages: C++, Python, Java, JavaScript, SQL (PostgreSQL and SQLite), Bash
Web: Flask, React, Django, HTML/CSS
Developer Tools: Git, Docker, Linux, VS Code, IntelliJ, Google Colab, Claude Code
Security: Nmap, Ghidra, BurpSuite
Libraries: TensorFlow, SkLearn, PyTorch, Pandas, Numpy, Matplotlib, SageMath, SciPy